

Wi-Fi FTM Versus UWB for 3D Indoor Positioning

Marios Raspopoulos
INSPIRE Research Centre
University of Central Lancashire
Larnaca, Cyprus
0000-0003-1513-6018

Iacovos Ioannou
CYENS Centre of Excellence
University of Cyprus
Nicosia, Cyprus
0000-0002-1562-5543

Nearchos Paspallis
INSPIRE Research Centre
University of Central Lancashire
Larnaca, Cyprus
0000-0002-2636-7973

Abstract—This paper evaluates Wi-Fi Fine Timing Measurement (FTM) for 3D indoor positioning and compares it with Ultra-Wideband (UWB). We follow a two-phase experimental methodology. First, we quantify ranging precision in a 45 m corridor under Line-of-Sight (LoS), Non-Line-of-Sight (NLoS), and mixed conditions. Second, we assess 3D positioning accuracy in a laboratory using nonlinear multilateration with four anchors, under both LoS and strong NLoS conditions. The evaluation uses commercial off-the-shelf FTM devices (Google Nest Wi-Fi and Pixel 6 Pro) and a Qorvo MDEK1001 UWB development kit. Results show that Wi-Fi FTM provides meter-level 3D positioning with minimal infrastructure, while UWB achieves substantially higher ranging precision and lower 3D positioning error under LoS. Under strong NLoS, both technologies suffer from reduced range availability, preventing 3D estimation when fewer than four ranges are returned. These results clarify the trade-offs between positioning accuracy, infrastructure requirements, and deployment complexity when selecting FTM or UWB for practical 3D indoor positioning.

Index Terms—3D indoor localization, Wi-Fi FTM, round-trip time (RTT), UWB

I. INTRODUCTION

Wi-Fi has been widely studied for indoor localization, often achieving room-level accuracy of approximately 2 m to 5 m using received signal strength (RSSI). However, RSSI-based accuracy often saturates due to multipath, environmental dynamics, and device variability. Wi-Fi Fine Timing Measurement (FTM) addresses these limitations by enabling time-of-flight-based ranging, rather than relying on signal strength or fingerprinting. In FTM, a mobile device measures the round-trip time (RTT) of exchanged frames with an access point and converts RTT into a distance estimate. When distances are collected from multiple FTM-capable access points, the 3D position of the target can be estimated using multilateration.

FTM is specified within the IEEE 802.11 family and enables two-way ranging using RTT measurements. More recent amendments extend these capabilities toward improved accuracy and wider bandwidth operation. IEEE 802.11az targets next-generation positioning and supports wider bandwidth operation for improved accuracy. IEEE 802.11bk further enhances FTM by enabling 320 MHz channel operation when compatible hardware is available. In practice, 802.11az and 802.11bk support is still limited in commercial devices, and

reported sub-meter performance is often demonstrated under controlled conditions and with careful calibration.

A key advantage of Wi-Fi FTM is that it can leverage existing Wi-Fi infrastructure and commercial off-the-shelf (COTS) devices that support IEEE 802.11 FTM. In contrast, Ultra-Wideband (UWB) and millimeter-wave sensing [1] can achieve higher precision but typically require specialized hardware and more constrained deployments. Although mmWave sensing can offer high precision, it can be limited by attenuation, sensitivity to obstructions, and sensitivity to orientation [2]. Fingerprinting approaches, while widely used, often require extensive offline calibration and frequent updates to maintain accuracy, which can be labor-intensive and sensitive to environmental changes [3].

Wi-Fi FTM can support practical 3D positioning for indoor navigation, asset tracking, and location-based services without requiring additional radio infrastructure beyond FTM-capable access points. This is attractive for environments such as hospitals, shopping malls, and industrial facilities. It may also enable applications such as augmented reality, autonomous robotics, and emergency response, where reliable location estimates improve efficiency and safety.

Contributions: (i) We provide a comparative experimental evaluation of commercial off-the-shelf (COTS) Wi-Fi FTM and UWB for 3D indoor positioning using identical anchor geometry and multilateration. (ii) We quantify how ranging precision affects 3D positioning accuracy under LoS and NLoS conditions. (iii) We report practical limitations under strong brick-wall NLoS conditions, where insufficient ranges prevent 3D estimation.

II. RELATED WORK

Various works have been proposed in the literature, leveraging the potential of Wi-Fi FTM to provide an easy-to-deploy positioning solution.

In [4], the authors use 802.11mc products (access points and smartphones) to set up a localization system based on round-trip time (RTT) measurements. They developed an algorithm that includes the characterization and removal of systematic biases to improve Wi-Fi RTT measurements, a clustering-based trilateration (CbT) approach with Weighted Concentric Circle Generation (WCCG) to handle non-intersecting range estimates, and a Kalman filter to refine position and trajectory

tracking. Experimental results in real-world indoor environments demonstrated an average ranging precision of 0.5 m at 80 MHz, an average static 2D positioning accuracy of approximately 1.2 m, and dynamic tracking accuracy of approximately 1.3 m. The method closely approaches the theoretical Cramer-Rao Bound (CRB), indicating high reliability. The authors of [5] have also tried to explore the potential of IEEE802.11mc for indoor localization. They demonstrated that Wi-Fi-RTT systematically underestimates distances and to account for this, they used a Gaussian Mixture Model (GMM) error model integrated into a Particle Filter (PF) localization framework. Their experiments were done using a Google Pixel 3 smartphone and Google Wi-Fi APs, in a controlled environment and an indoor setting with 6 access points. Results indicate a consistent negative bias of approximately $-1.3m$ in range measurements. Their proposed methodology compensates for this bias and improves positioning accuracy to 0.47 m.

In [6], the author compares the two-sided RTT ranging approach supported by 802.11mc-capable access points with one-sided RTT (supported in Android 12 and later) that operates with legacy access points. The results indicate that one-sided RTT achieves 3–4m 2D positioning accuracy in challenging indoor conditions under a 20MHz channel, whereas the two-sided approach achieves a 1–2m accuracy at an 80MHz channel. These results suggest that one-sided RTT can serve as a viable alternative in settings where upgrading access points to IEEE 802.11 FTM is not feasible. However, one disadvantage is that the access point turn-around time, which must be compensated, is not known a priori. Like the previous work, this one also examines RTT measurement biases, methods to estimate them efficiently, and techniques for selecting optimal APs to improve accuracy.

The work of [7] compares Wi-Fi FTM with Wi-Fi RSSI methods in realistic indoor environments using different position estimation methods. The results confirm that FTM provides more accurate distance measurements than RSSI, but its error behaviour still exhibits similarities to RSSI-based methods, especially in environments with non-line-of-sight (NLOS) propagation.

The authors of [8] follow a slightly different approach. They combine IEEE802.11mc RTT measurements and fingerprinting techniques. While traditional RSS-based fingerprinting is subject to environmental variability and the fact that RTT-based multilateration might be prone to errors due to multipath and non-line-of-sight conditions, they proposed a system called WiNar that integrates RTT with RSS in a hybrid fingerprinting approach, leveraging commercial off-the-shelf (COTS) 802.11mc devices. The system does not require clock synchronization and enhances accuracy using a novel weighted similarity model for RTT-RSS fingerprinting. Their results indicate that WiNar achieves sub-meter accuracy, with a mean localization error of 0.73m in a campus testbed and 0.77m in an office testbed.

A machine learning (ML) approach is proposed in [9]. The work focuses on the problem of inaccurate Wi-Fi RTT measurements and proposes a neural network (NN) model

that adaptively learns measurement patterns across different indoor environments, producing more accurate ranging results. It also incorporates Pedestrian Dead Reckoning (PDR) from smartphone sensors to estimate a reference trajectory, which is used to train the NN model without requiring ground truth labels. Experimental results from an indoor office environment with 10 Wi-Fi FTM access points indicate an average accuracy of 1.24m for a 40MHz bandwidth scenario. The findings suggest that sensor fusion with unsupervised learning can significantly improve Wi-Fi RTT-based indoor localization, making it a viable solution for real-world applications.

Despite the potential of Wi-Fi FTM for indoor localization, several works raise security and privacy concerns. For instance, the works reported in [10], [11]. Both studies suggest that Wi-Fi FTM needs cryptographic security, better MAC address randomization, and passive positioning techniques to improve security and privacy in future Wi-Fi positioning systems, given that FTM messages are unencrypted.

Overall, the general takeaway from this review is that IEEE 802.11mc FTM significantly enhances Wi-Fi-based indoor positioning, achieving meter-level accuracy. However, challenges such as multipath interference, hardware variability, and security remain issues to solve. In this work, we evaluate the ranging precision of Wi-Fi of COTS 802.11mc devices and compare it to that of UWB for 3D positioning. UWB has proven to be a well-established technology for localization but its applicability is questionable given the specific infrastructure requirements. Several systems were implemented commercially, while many others are being utilised in experimental testbeds such as those provided by Decawave and Bespoon. These systems have been thoroughly researched and validated for specific purposes. Other activities focused on the Non-Line-of-Sight (NLOS) problem, which is a primary source of inaccuracy in UWB ranging and positioning [12]. The authors of [13] propose a UWB positioning system which utilizes a two-way-time-of-flight (TWTF) for range measurements, which, when used in a multilateration estimation algorithm, yielded an average 3D accuracy of $100 \pm 25mm$. In [14], the authors propose a 3D ToA algorithm using UWB in which they replaced the quadratic term in the positioning estimation with a new variable using the weighted least squares linear estimation followed by a Kalman filter. The simulation results indicated that the positioning accuracy can reach 5–10cm. [15] presents a novel UWB self-localizing anchor-system calibration approach that uses a calibration unit (CU) for improved localization accuracy and confirmed that using the CU decreases the positional error of the anchors in 3D UWB localization systems, hence achieving better positioning accuracy of around 0.32m.

While prior studies have evaluated Wi-Fi FTM or UWB independently, direct experimental comparisons between COTS FTM and UWB systems for 3D positioning under identical conditions remain limited. This work addresses this gap through a controlled comparative evaluation using identical anchor geometry and positioning methodology.



Fig. 1. Precision Analysis Environment

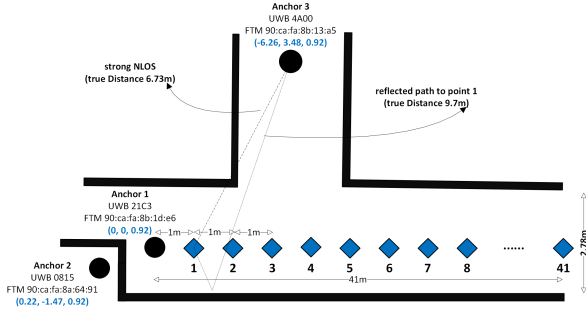


Fig. 2. Precision Analysis Setup

III. METHODOLOGY

A. System Overview

As mentioned earlier, this study evaluates FTM-based technology for 3D positioning and compares its performance with the well-established UWB technology. The methodology for this study includes a precision analysis experiment using off-the-shelf FTM-enabled devices and a UWB kit to evaluate and compare the precision of these technologies under Line-of-Sight (LoS) and Non-Line-of-Sight (NLoS) conditions. Then, a cluster of anchors for each technology is deployed in an indoor environment to evaluate and compare 3D positioning accuracy using a nonlinear 3D multilateration approach, under both LoS and NLoS conditions.

B. Hardware and Software

For the FTM experimentation, we used the Google Nest Wi-Fi access points that support 802.11mc together with a Google Pixel 6 Pro as a target. We developed a custom Android application for both the precision analysis and the positioning accuracy experiments using the `android.net.wifi.rtt` API. For the UWB experimentation, we used the Qorvo MDEK1001 Development Kit, a comprehensive platform consisting of 12 nodes for developing and evaluating UWB positioning systems. The kit is based on the Decawave DW1000 transceiver, which operates in the 3.5-6.5 GHz frequency range with a bandwidth of 500 MHz. Each node can be configured as either an anchor or a tag, allowing flexible

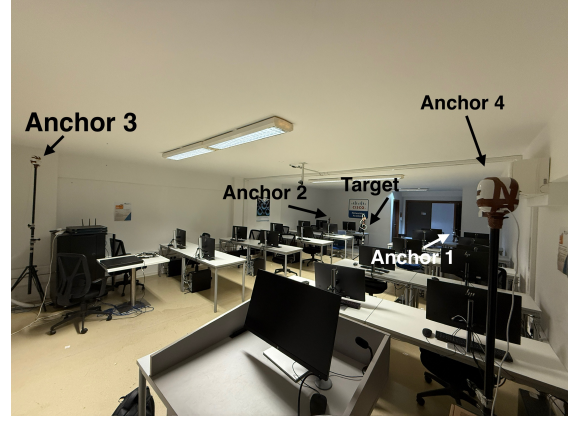


Fig. 3. Positioning Experiments Environment

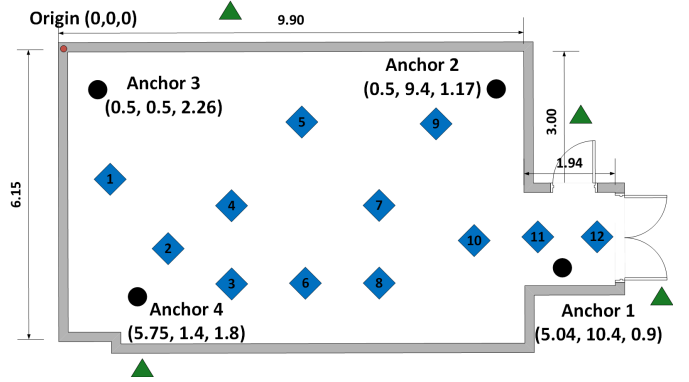


Fig. 4. Positioning Experiments Setup (x-axis increases downwards, y-axis increases to the right)

deployment for positioning experiments. Although the kit includes an Android-based application with a manufacturer-claimed 2D positioning accuracy of $<10\text{cm}$, this application does not support real-time acquisition of raw range measurements. To overcome this limitation, we developed a Python implementation that leverages the kit API and serial communication for direct data acquisition and processing.

The experiments were conducted in an operational university environment with multiple active Wi-Fi networks. The Google Nest Wi-Fi access points operated in the 5 GHz band using IEEE 802.11mc FTM. No channel isolation or interference mitigation was applied to evaluate performance under realistic deployment conditions. Consequently, the reported FTM results inherently include the impact of ambient Wi-Fi interference and channel contention.

C. Experimental Setup

This section outlines the experimental setup used to evaluate both systems' precision and 3D positioning accuracy.

1) *Precision Analysis*: The precision analysis was conducted to compare the ranging accuracy capabilities of the FTM and UWB systems. Measurements were conducted along a 45 m corridor, as shown in Figure 1. 3 anchors were set up

in the locations indicated with black solid circles in Figure 2 to emulate a pure LoS (Anchor 1) measurement series, a pure NLoS (Anchor 2) one, and a mixed one (Anchor 3). Range measurements were collected from every anchor along a straight line consisting of equally spaced (1m) points depicted with blue diamonds in Figure 2. At each measurement point, 20 range measurements were collected at 10 samples/s, and we report the mean and standard deviation.

2) *3D Positioning Accuracy*: The 3D positioning experiments were conducted in a 6.3 m × 11.84 m laboratory with a ceiling height of 2.5 m, as shown in Figures 3 and 4. The setup included four anchors—each comprising an FTM access point and a UWB module mounted together on tripods—placed at the four corners of the lab, indicated by solid black circles in Figure 4. Measurements were taken at 12 different locations throughout the room, marked by blue diamonds in Figure 4, and at varying heights. At each location, 20 measurements were collected. For each anchor at each location, we remove outliers defined as measurements more than three standard deviations from the local mean, and then compute the average range. To evaluate the positioning performance under NLoS conditions, we moved the 4 anchors outside the room at locations marked with green triangles in Figure 4. 3D position estimation was performed using a non-linear multilateration formulation.

3) *Nonlinear 3D Multilateration*: Let the unknown position of a target be denoted by the vector $\mathbf{p} = (x, y, z)^T \in \mathbb{R}^3$. Assume that there are $N \geq 4$ anchor nodes with known positions $\mathbf{a}_i = (x_i, y_i, z_i)^T \in \mathbb{R}^3$ for $i = 1, \dots, N$. The predicted distance from the i -th anchor is:

$$\hat{d}_i = \|\mathbf{p} - \mathbf{a}_i\|_2 = \sqrt{(x - x_i)^2 + (y - y_i)^2 + (z - z_i)^2}.$$

Due to measurement noise, the measured distance is modeled as $d_i = \hat{d}_i + \varepsilon_i$, where ε_i is a random error term.

To estimate the unknown position $\mathbf{p}$, we formulate a nonlinear least squares problem that minimizes the residuals between the measured and computed distances:

$$\min_{\mathbf{p} \in \mathbb{R}^3} \sum_{i=1}^N (\|\mathbf{p} - \mathbf{a}_i\|_2 - d_i)^2$$

This optimization problem is nonlinear due to the Euclidean norm and is typically solved using iterative methods such as the Levenberg–Marquardt algorithm or Trust Region Reflective methods. The objective function can be rewritten as:

$$f(\mathbf{p}) = \sum_{i=1}^N \left(\sqrt{(x - x_i)^2 + (y - y_i)^2 + (z - z_i)^2} - d_i \right)^2$$

Solving this yields an estimate $\hat{\mathbf{p}}$ that best fits the measured distances in a least-squares sense. An initial guess is needed, which is assumed to be the mean of the anchor locations.

IV. RESULTS

1) *Precision Analysis*: The precision analysis results for the FTM and UWB setups are presented in Figures 5 and 6, respectively. In both cases, the solid blue line represents the

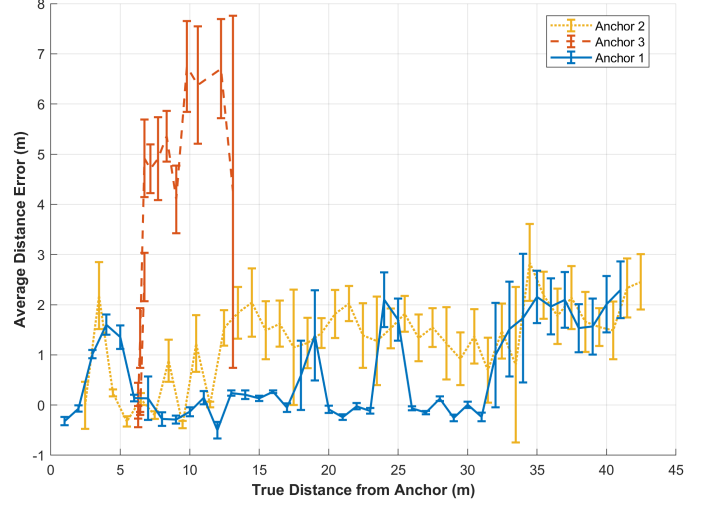


Fig. 5. FTM Precision Analysis Results

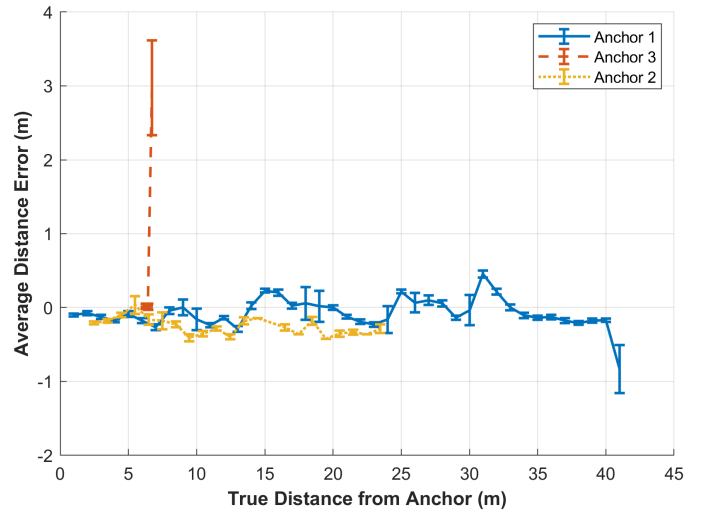


Fig. 6. UWB Precision Analysis Results

distance error along the route under the LoS (Line-of-Sight) condition for Anchor 1. For LoS (Anchor 1), the FTM setup achieved a mean error of 0.63 m with a standard deviation of 0.90 m, while the UWB setup achieved a mean error of -0.07 m with a standard deviation of 0.20 m over the full route. In the NLoS (Non-Line-of-Sight) case, corresponding to Anchor 2 and illustrated with a dotted orange line in both figures, the FTM setup exhibited a mean error of 1.31 m with a standard deviation of 0.78 m. In contrast, the UWB setup reported a mean error of 0.25 m with a standard deviation of 0.11 m. Clearly, the UWB sensors demonstrated a significantly better ranging precision in both LoS and NLoS conditions, both in terms of mean error and variability. However, the FTM setup managed to record range measurements across the entire route in the NLoS scenario, albeit with higher ranging errors. On the other hand, the UWB system was only able to return measurements up to a maximum distance of 24m due to the strong attenuation caused by the brick wall. The

TABLE I
COMPARISON OF FTM AND UWB POSITIONING ERRORS UNDER LoS CONDITIONS

Point	True Position			Dilution of Precision			FTM Positioning					UWB Positioning				
ID	X	Y	Z	HDOP	VDOP	PDOP	Accum. Dist Error	Dx	Dy	Dz	XYZ Error	Accum. Dist Error	Dx	Dy	Dz	XYZ Error
1	2.13	0.80	1.75	1.22	4.48	4.64	2.41	0.24	0.69	0.42	0.84	1.53	0.24	0.20	1.00	1.05
2	4.51	1.92	2.04	1.40	6.29	6.45	6.59	0.80	1.47	0.45	2.22	2.72	0.17	0.07	0.38	0.24
3	5.07	3.40	1.15	1.42	4.32	4.55	2.81	1.83	1.24	0.22	1.98	1.47	0.07	0.01	0.23	0.51
4	3.14	3.37	1.50	1.61	9.02	9.17	1.25	1.30	0.31	1.47	0.46	1.22	0.29	0.09	0.41	0.23
5	1.07	5.02	1.40	1.64	10.29	10.42	3.48	0.99	2.33	1.50	1.27	1.71	0.18	0.10	0.49	0.46
6	4.81	4.80	1.63	2.37	18.60	18.75	1.90	0.14	0.39	0.21	0.66	0.72	0.04	0.07	0.22	0.18
7	3.30	6.50	1.67	1.59	8.79	8.94	2.05	0.02	0.45	2.15	0.44	1.46	0.28	0.18	0.43	0.31
8	5.05	6.45	1.55	1.78	12.30	12.43	2.36	0.67	0.79	0.74	1.00	1.90	0.40	0.03	0.23	0.37
9	1.37	8.02	1.05	1.23	6.44	6.56	2.61	1.29	0.02	1.74	1.51	2.34	0.48	0.10	0.24	0.39
10	3.35	8.70	1.88	1.12	2.57	2.80	2.15	0.60	0.25	0.14	1.13	1.58	0.02	0.05	0.17	0.90
11	4.20	9.70	1.10	1.92	12.07	12.23	5.59	0.10	1.69	2.21	2.34	0.57	0.07	0.14	0.25	0.24
12	4.16	11.33	1.72	1.23	2.14	2.47	5.15	2.14	0.16	0.28	2.48	2.00	0.07	0.12	0.23	0.77
Average							3.19	0.84	0.82	0.96	1.36	1.60	0.19	0.10	0.36	0.47

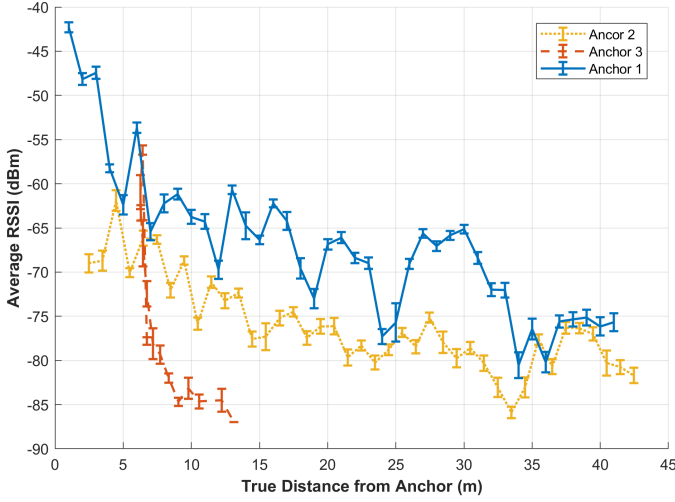


Fig. 7. FTM RSSI plot along the precision analysis route

observed FTM fluctuations may be due to multipath and, potentially, interference from other Wi-Fi systems operating on overlapping channels. In the mixed scenario (Anchor 3), indicated by the red dashed line in both figures, reliable measurements were obtained only for the LoS points (points 2 to 5). Measurements at points obstructed by brick-wall corners either showed high error or did not return results. The elevated error is likely due to severe attenuation of the LoS component, with the ranging result being derived from a reflected signal, most likely from the corridor wall. This hypothesis is supported by the geometrical illustration for point 1 in Figure 2, which shows that the reflected path is approximately 3 m longer than the direct LoS path. To illustrate the degree of attenuation caused by the brick wall in the NLoS cases, the RSSI measurements collected using the FTM setup are presented in Figure 7.

2) *3D Positioning Accuracy*: The 3D positioning results, obtained using nonlinear multilateration and the setup described in Section III-C, are shown in Table I. The table includes the ground truth of the 12 measurement points as well as the Dilution of Precision (DOP) values. DOP is a numerical

measure that describes how the geometry or constellation of anchors affects the accuracy of a position estimate. The DOP values are derived from the geometry matrix G , which consists of unit vectors pointing from the receiver to each anchor. For a set of N anchors with known positions (x_i, y_i, z_i) and a receiver at position (x, y, z) , each row of G is defined as $\left[\frac{x-x_i}{r_i}, \frac{y-y_i}{r_i}, \frac{z-z_i}{r_i}, 1\right]$, where r_i is the distance between the receiver and the i -th anchor. The covariance matrix of the position estimate is given by $Q = (G^T G)^{-1}$, from which the DOP values are computed. The Position DOP (PDOP) is defined as $PDOP = \sqrt{q_{11} + q_{22} + q_{33}}$, where q_{ii} are the diagonal elements of Q corresponding to the variances in the x , y , and z directions. Similarly, the Horizontal DOP (HDOP) and Vertical DOP (VDOP) are given by $HDOP = \sqrt{q_{11} + q_{22}}$ and $VDOP = \sqrt{q_{33}}$, respectively. Lower DOP values indicate a better geometric configuration and lead to higher accuracy in position estimation. Typically, high HDOP is associated with larger errors in the x and y dimensions, while high VDOP is associated with larger errors in the z dimension. The table also presents the positioning accuracy results for each technology in relation to the range measurement precision under LoS conditions. Specifically, for each system, it includes the accumulated range error at each point, calculated as the sum of the absolute differences between the true distance and the measured distance from each anchor $\sum_{i=1}^N |d_i^{\text{meas}} - d_i^{\text{true}}|$ where d_i^{meas} is the measured distance and d_i^{true} is the ground-truth distance to anchor i , and N is the number of anchors. The table also reports the absolute positioning error along each axis: $\Delta x = |x^{\text{est}} - x^{\text{true}}|$, $\Delta y = |y^{\text{est}} - y^{\text{true}}|$, $\Delta z = |z^{\text{est}} - z^{\text{true}}|$ as well as the 3D Euclidean positioning error, computed as $\sqrt{(x^{\text{est}} - x^{\text{true}})^2 + (y^{\text{est}} - y^{\text{true}})^2 + (z^{\text{est}} - z^{\text{true}})^2}$ where $(x^{\text{est}}, y^{\text{est}}, z^{\text{est}})$ is the estimated position, and $(x^{\text{true}}, y^{\text{true}}, z^{\text{true}})$ is the ground-truth location.

The results indicate superior positioning accuracy for UWB, primarily due to its more precise and consistent range measurements. Specifically, the UWB system achieved an average accumulated range error of 1.60 m and an average 3D positioning error of 0.47 m. In contrast, the FTM system achieved an average accumulated range error of 3.19 m and

an average 3D positioning error of 1.36 m. The UWB system also demonstrated more stable positioning performance, with most errors remaining low, except in a few instances where they reached 1m — primarily due to high Dilution of Precision (DOP). In contrast, the FTM system showed greater variability, with positioning errors ranging from 0.44m to 2.48m, largely attributed to lower ranging precision. Both systems exhibited poorer performance along the z-axis, which can be attributed to the limited vertical distribution of the anchors, resulting in a high VDOP.

To evaluate the performance of both systems under non-line-of-sight (NLoS) conditions, we repositioned all access points outside the lab, placing them at the locations marked by green triangles in Figure 4. Across all 12 test locations, neither system produced a valid 3D position estimate because it was not possible to obtain ranges from all four anchors at each location. At best, specifically at locations 4 and 7, which are more centrally located within the lab, three ranges were recorded. However, this was still insufficient to estimate a 3D position. Results highlight the limitations of both technologies when operating under strong NLoS conditions. It is worth noting that the lab is enclosed by 20 cm-thick brick walls.

V. DISCUSSION & CONCLUSION

In this paper, we evaluated the potential of Wi-Fi FTM for 3D indoor localization and compared it with Ultra-Wideband (UWB). Our evaluation involved both a ranging precision analysis and a 3D indoor localization experiment using a non-linear multilateration approach.

While FTM showed reasonably good performance compared to other Wi-Fi-based localization methods, achieving an average 3D positioning error of approximately 1.36 m, it was less accurate than the UWB system, which achieved sub-meter accuracy (0.47 m). This difference is primarily due to the lower precision and higher variability of the FTM range measurements, as demonstrated in the precision analysis. This lower precision can be attributed to several factors: (1) narrower bandwidth, as IEEE 802.11mc operates on an 80MHz channel compared to UWB's minimum of 500MHz—though future standards such as IEEE 802.11az (160MHz) and IEEE 802.11bk (320MHz) are expected to improve this; (2) lower time resolution and less accurate clock synchronization, since standard Wi-Fi chipsets typically lack the high-resolution time-keeping and clock alignment mechanisms that UWB positioning systems are designed explicitly for; (3) increased multipath sensitivity, as narrowband systems like Wi-Fi are more affected by multipath propagation than UWB; (4) Wi-Fi interference, given that Wi-Fi bands are more congested and less robust to noise compared to the relatively less crowded UWB spectrum for these systems; and (5) hardware and calibration constraints, as FTM relies on consumer-grade Wi-Fi chipsets not optimized for precise ranging, whereas UWB hardware is purpose-built for positioning and typically includes calibrated components, dedicated antennas, and optimized algorithms.

Finally, both systems are substantially more effective under LoS conditions. Our precision analysis showed that non-line-

of-sight (NLoS) environments significantly affect their performance, either in range accuracy for UWB or in precision for FTM. This was reflected in our NLoS positioning experiments, where neither system produced valid position estimates due to the lack of sufficient range measurements.

Beyond accuracy, system deployability is an important consideration. UWB provides higher positioning precision but requires dedicated infrastructure and specialized hardware. In contrast, Wi-Fi FTM can leverage existing Wi-Fi infrastructure and commodity devices, reducing deployment cost and complexity. Therefore, UWB is preferable for precision-critical applications, whereas FTM may offer a better cost-performance trade-off where meter-level accuracy is sufficient.

REFERENCES

- [1] A. Sesyuk, S. Ioannou, and M. Raspopoulos, "3d millimeter-wave sensing vs ultra-wideband positioning," in *2024 14th International Conference on Indoor Positioning and Indoor Navigation (IPIN)*, 2024, pp. 1–6.
- [2] —, "Radar-based millimeter-wave sensing for accurate 3-d indoor positioning: Potentials and challenges," *IEEE Journal of Indoor and Seamless Positioning and Navigation*, vol. 2, pp. 61–75, 2024.
- [3] M. Raspopoulos, "Multidevice map-constrained fingerprint-based indoor positioning using 3-d ray tracing," *IEEE Transactions on Instrumentation and Measurement*, vol. 67, no. 2, pp. 466–476, 2018.
- [4] C. Ma, B. Wu, S. Poslad, and D. R. Selviah, "Wi-fi rtt ranging performance characterization and positioning system design," *IEEE Transactions on Mobile Computing*, vol. 21, no. 2, pp. 740–756, 2022.
- [5] C. Gentner, M. Ulmschneider, I. Kuehner, and A. Dammann, "Wifi-rtt indoor positioning," in *2020 IEEE/ION Position, Location and Navigation Symposium (PLANS)*, 2020, pp. 1029–1035.
- [6] B. K. P. Horn, "Indoor localization using uncooperative wi-fi access points," *Sensors*, vol. 22, no. 8, 2022. [Online]. Available: <https://www.mdpi.com/1424-8220/22/8/3091>
- [7] M. Bullmann, T. Fetzter, F. Ebner, M. Ebner, F. Deinzer, and M. Grzegorzec, "Comparison of 2.4 ghz wifi ftm- and rssi-based indoor positioning methods in realistic scenarios," *Sensors*, vol. 20, no. 16, 2020. [Online]. Available: <https://www.mdpi.com/1424-8220/20/16/4515>
- [8] O. Hashem, K. A. Harras, and M. Youssef, "Accurate indoor positioning using ieee 802.11mc round trip time," *Pervasive and Mobile Computing*, vol. 75, p. 101416, 2021. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1574119221000687>
- [9] J. Choi, "Enhanced wi-fi rtt ranging: A sensor-aided learning approach," *IEEE Transactions on Vehicular Technology*, vol. 71, no. 4, pp. 4428–4437, 2022.
- [10] G. Singh, A. Pandey, M. Prakash, M. Andreoni, and M. Baddeley, "Benchmarking and security considerations of wi-fi ftm for ranging in iot devices," in *Proceedings of Cyber-Physical Systems and Internet of Things Week 2023*, ser. CPS-IoT Week '23. New York, NY, USA: Association for Computing Machinery, 2023, p. 67–71. [Online]. Available: <https://doi.org/10.1145/3576914.3587520>
- [11] S. Schepers and A. Ranganathan, "Privacy-preserving positioning in wi-fi fine timing measurement," *Proceedings on Privacy Enhancing Technologies*, vol. 2, p. 325–343, 2022.
- [12] A. Jiménez and F. Seco, "Comparing decawave and bespoon uwb location systems: Indoor/outdoor performance analysis," in *2016 International Conference on Indoor Positioning and Indoor Navigation (IPIN)*, 2016, pp. 1–8.
- [13] P. Dabove, V. Di Pietra, M. Piras, A. A. Jabbar, and S. A. Kazim, "Indoor positioning using ultra-wide band (uwb) technologies: Positioning accuracies and sensors' performances," in *2018 IEEE/ION Position, Location and Navigation Symposium (PLANS)*, 2018, pp. 175–184.
- [14] D. Ni, O. A. Postolache, C. Mi, M. Zhong, and Y. Wang, "Uwb indoor positioning application based on kalman filter and 3-d toa localization algorithm," in *2019 11th International Symposium on Advanced Topics in Electrical Engineering (ATEE)*, 2019, pp. 1–6.
- [15] P. Krapež and M. Munih, "Anchor calibration for real-time-measurement localization systems," *IEEE Transactions on Instrumentation and Measurement*, vol. 69, no. 12, pp. 9907–9917, 2020.